

Multiple Imputation Strategy for Quality of Life (QoL) Outcome

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Overview

This document describes the multiple imputation strategy for the Quality of Life (QoL) outcome in the LIQPLAT trial. The approach uses Multiple Imputation by Chained Equations (MICE) with multilevel predictive mean matching (PMM) to handle missing data under the Missing At Random (MAR) assumption.

Setup

```
library(here)
library(tidyverse)
library(patchwork)
library(mice)
library(miceadds)
library(arrow)
```

Data

The following code uses historical data from the Division of Medical Oncology at USB (first appointments from 2023 to mid-2025, prior to LIQPLAT initiation). These data are used to develop and validate the imputation procedure; actual trial data are not publicly available.

```
set.seed(1234)
df_raw <- read_parquet(here("data", "alt_qol.parquet"))
```

Data Preprocessing

Inclusion Criteria

The substantive analysis model includes only patients who survived beyond 182 days. To maintain congruence between the imputation and analysis models, we restrict the imputation dataset to:

1. Observations up to day 182 after randomisation
2. Patients who either have no recorded death or died after day 182

Variable Transformations

Ordered factors (ECOG, QoL, distress) are converted to numeric format for imputation. This avoids sparse dummy variable matrices that can cause convergence issues in MICE. Predictive mean matching preserves the discrete nature of these variables by imputing only observed values.

```
tx <- sample(unique(df_raw$pat_id), size = 0.5 * length(unique(df_raw$pat_id)), replace
= FALSE)

imputation_vars <- c(
  "pat_id",
  "tx",
  "gender",
  "age",
  "ecog_fstcnt",
  "diagnosis",
  "plan_fstcnt_coded",
  "week",
  "distress",
  "q30"
  # Variables available in trial data but not in historical dataset:
  # "albumin",
  # "c_reactive_protein"
)

df <- df_raw |>
  filter(quest_day <= 182, (is.na(death_day) | death_day >= 182)) |>
  mutate(
    tx = if_else(pat_id %in% tx, "1", "0"),
    distress = as.numeric(as.factor(coalesce(distress_therm_2, distress_therm_v2,
distress_therm_inpat))), ordered = TRUE),
    week = ceiling(quest_day / 7),
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    plan_fstcnt_coded = fct_lump_n(plan_fstcnt_coded, n = 4),
    plan_fstcnt_coded = factor(as.integer(plan_fstcnt_coded)),
    q30 = as.numeric(as.character(q30)),
    ecog_fstcnt = as.numeric(as.character(ecog_fstcnt)),
    tx = factor(tx)
  ) |>
  rename(age = pat_age) |>
  select(all_of(imputation_vars)) |>
  arrange(pat_id, week)

# Extract baseline QoL: last observation at or before random selection (week <= 0)
baseline_qol <- df |>
  filter(week <= 0) |>
```

```

select(pat_id, q30, week) |>
group_by(pat_id) |>
arrange(desc(week)) |>
slice_head(n = 1) |>
ungroup() |>
select(pat_id, baseline_q30 = q30)

df <- df |>
left_join(baseline_qol, by = "pat_id") |>
filter(week > 0) |>
relocate(baseline_q30, ecog_fstcnt, diagnosis, plan_fstcnt_coded, .after = pat_id)

# Model non-linear time effects using restricted cubic splines with 4 knots
basis_rcs <- rms::rcs(df$week, 4)

df$week <- as.vector(basis_rcs[, 1])
df$week_nl1 <- as.vector(basis_rcs[, 2])
df$week_nl2 <- as.vector(basis_rcs[, 3])

```

Imputation Variables

The imputation model includes variables from the substantive analysis model and auxiliary variables that are predictive of missing values or their missingness mechanism [Mathur2025-sl; van-Buuren2021-pc].

Analysis Variables

Variable	Description	Level
tx	Treatment group allocation	Baseline
age	Age at date of randomisation (years)	Baseline
gender	Patient sex	Baseline
ecog_fstcnt	ECOG performance status at baseline (0–4)	Baseline
diagnosis	Primary cancer diagnosis (5 categories)	Baseline
baseline_q30	EORTC QLQ-C30 Q30 at baseline	Baseline
week	Week of questionnaire completion (post-randomisation)	Longitudinal
q30	EORTC QLQ-C30 question 30 (overall QoL, 1–7)	Longitudinal

Table 1: Analysis variables included in the imputation model

Auxiliary Variables

Variable	Description	Level
distress	Distress thermometer score (0–10)	Longitudinal
plan_fstcnt_coded	Treatment intent at baseline (5 categories)	Baseline
albumin	Serum albumin at baseline (g/L)	Baseline
c_reactive_protein	C-reactive protein at baseline (mg/L)	Baseline

Table 2: Auxiliary variables included in the imputation model. Albumin and CRP are available in trial data but not in the historical dataset used here.

Multiple Imputation Procedure

We employ multilevel multiple imputation to account for the hierarchical data structure, where repeated QoL measurements (Level 1) are nested within patients (Level 2). The patient identifier (`pat_id`) serves as the clustering variable.

Imputation Methods

Baseline variables (Level 2) are imputed using `2lonly.pmm` from the `miceadds` package. This method:

1. Aggregates longitudinal predictors to the cluster level by computing patient means
2. Fits a linear regression model on the collapsed data:

$$Y_j = \beta_0 + \beta_1 \mathbf{X}_j + \beta_2 \bar{\mathbf{Z}}_j + \epsilon_j$$

where Y_j is the missing baseline variable for patient j , $\mathbf{X}_j$ represents other baseline predictors, and $\bar{\mathbf{Z}}_j$ represents aggregated means of longitudinal predictors

3. Selects imputed values via predictive mean matching

Longitudinal variables (Level 1) are imputed using `2l.pmm`, which fits a linear mixed-effects model:

$$Y_{ij} = \beta_0 + \beta_1 \mathbf{X}_{ij} + u_{0j} + \epsilon_{ij}$$

where Y_{ij} is the outcome for observation i of patient j , $\mathbf{X}_{ij}$ includes fixed effects (time-varying and static predictors), and $u_{0j} \sim N(0, \sigma_u^2)$ is the patient-specific random intercept.

For both methods, we use Type 1 predictive mean matching with a donor pool of $k = 5$ [van-Buuren2021-pc, chapters 3.4, 7.3, 7.8].

Predictor Matrix Configuration

The predictor matrix is configured to:

- Use `pat_id` as the clustering variable (coded as `-2`)
- Prevent time variables from predicting static baseline variables
- Allow baseline variables to predict longitudinal outcomes

```
md.pattern(df)
```

```
init <- mice(df, maxit = 0)
pred_matrix <- init$predictorMatrix
meth <- init$method

# Set patient ID as clustering variable
pred_matrix[, "pat_id"] <- -2
pred_matrix["pat_id", ] <- 0

# Time variables should not predict static baseline variables
baseline_vars <- c("tx", "age", "gender", "diagnosis", "ecog_fstcnt",
  "plan_fstcnt_coded", "baseline_q30")
pred_matrix[c(baseline_vars, "week", "week_n11", "week_n12"), c("week", "week_n11",
  "week_n12")] <- 0

# Longitudinal variables: multilevel PMM
meth["q30"] <- "2l.pmm"
meth["distress"] <- "2l.pmm"
```

```
# Baseline variables: cluster-level PMM
meth["ecog_fstcnt"] <- "2lonly.pmm"
meth["plan_fstcnt_coded"] <- "2lonly.pmm"
meth["diagnosis"] <- "2lonly.pmm"
meth["baseline_q30"] <- "2lonly.pmm"

imputed_data <- mice(df,
  m = 5,          # 50 in final analysis
  maxit = 5,      # 50 in final analysis
  predictorMatrix = pred_matrix,
  method = meth,
  donors = 5,
  seed = 78109823)
```

Imputation Parameters

For the final trial analysis:

- Number of imputed datasets: $m = 50$
- Maximum iterations: 50
- Donor pool size: $k = 5$

Convergence Diagnostics

Convergence is assessed via trace plots of the mean and standard deviation of imputed values across iterations.

```
plot(imputed_data)
```

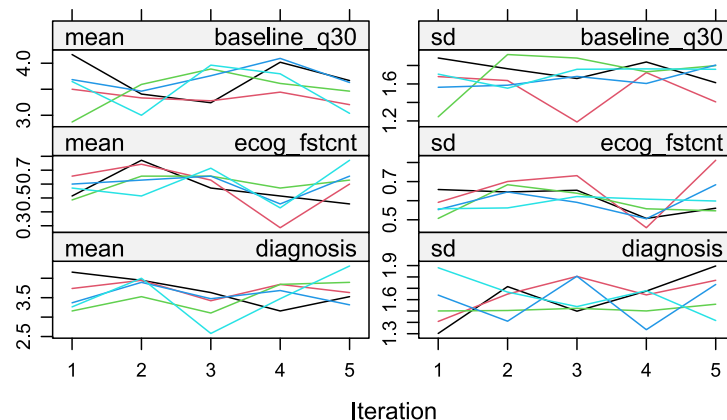


Figure 1: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

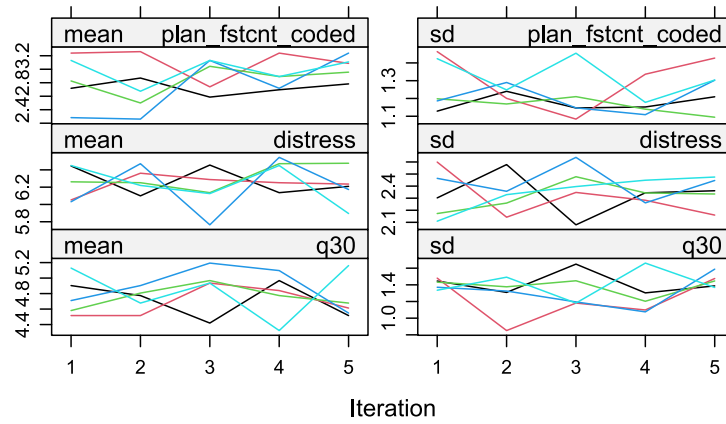


Figure 2: Trace plots showing the mean and standard deviation of imputed values across iterations for each imputed dataset.

Distribution Diagnostics

We compare the distributions of observed and imputed values to assess plausibility.

```
densityplot(imputed_data, ~baseline_q30 + q30 + ecog_fstcnt + diagnosis +
plan_fstcnt_coded)
```

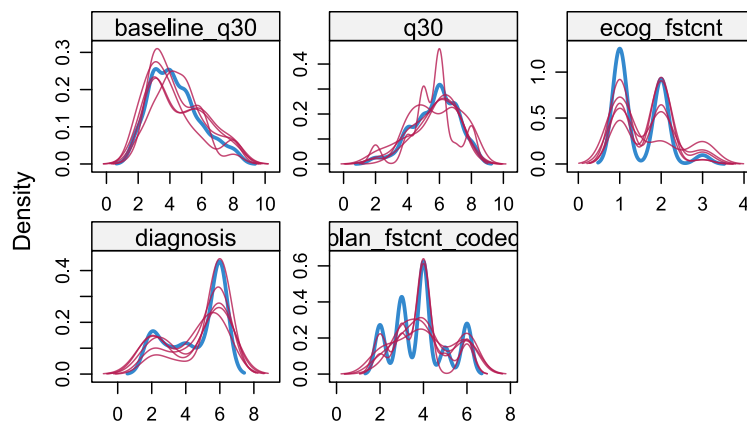


Figure 3: Density plots comparing observed (blue) and imputed (red) values for baseline variables and QoL.

```
stripplot(imputed_data, baseline_q30 + ecog_fstcnt + diagnosis + plan_fstcnt_coded
~ .imp)
```

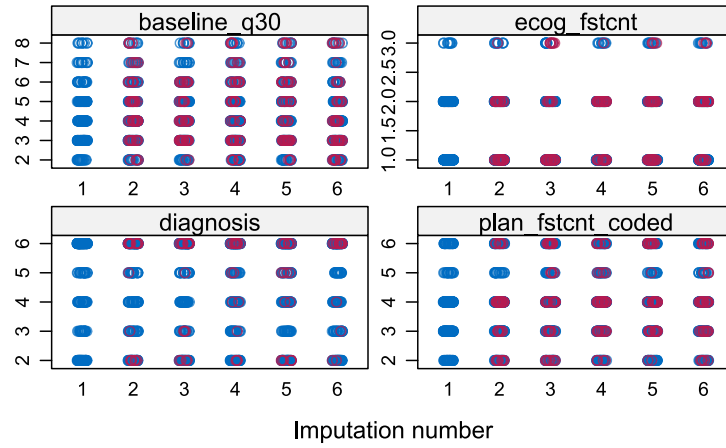


Figure 4: Strip plots of imputed values for baseline variables across imputed datasets.

```
xyplot(imputed_data, q30 ~ week | .imp, pch = c(1, 20))
```

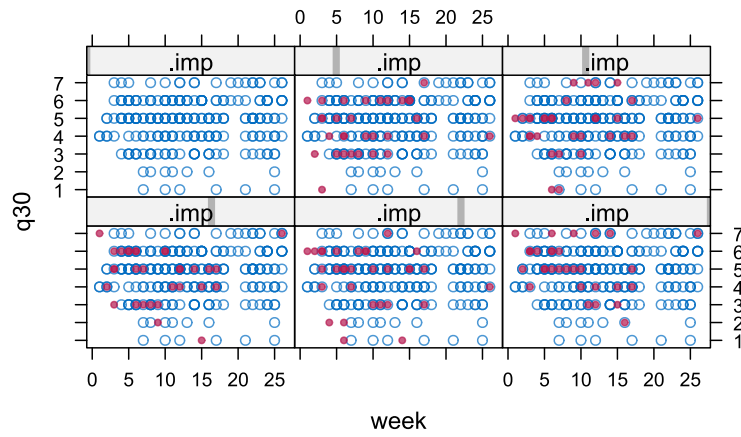


Figure 5: Scatter plots of QoL against time, by imputed dataset and treatment group, and against baseline QoL.

```
xyplot(imputed_data, q30 ~ week | tx, pch = c(1, 20))
```

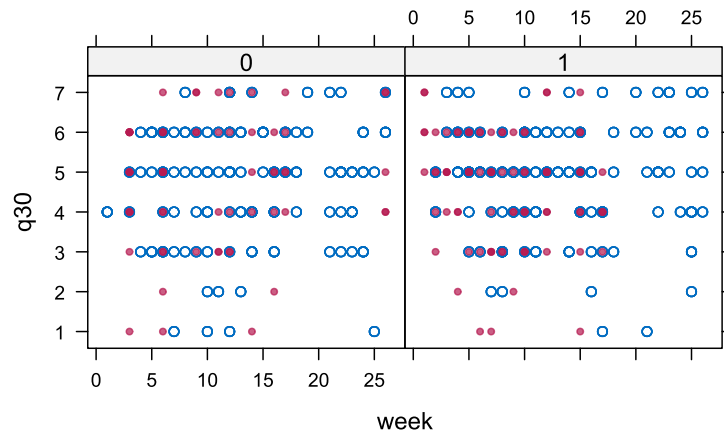


Figure 6: Scatter plots of QoL against time, by imputed dataset and treatment group, and against baseline QoL.

```
xyplot(imputed_data, q30 ~ baseline_q30 | .imp, pch = c(1, 20))
```

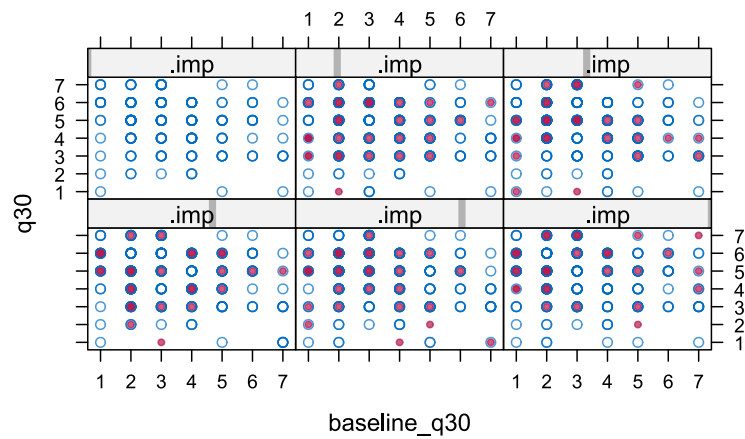


Figure 7: Scatter plots of QoL against time, by imputed dataset and treatment group, and against baseline QoL.

```
xyplot(imputed_data, q30 ~ distress | .imp, pch = c(1, 20))
```

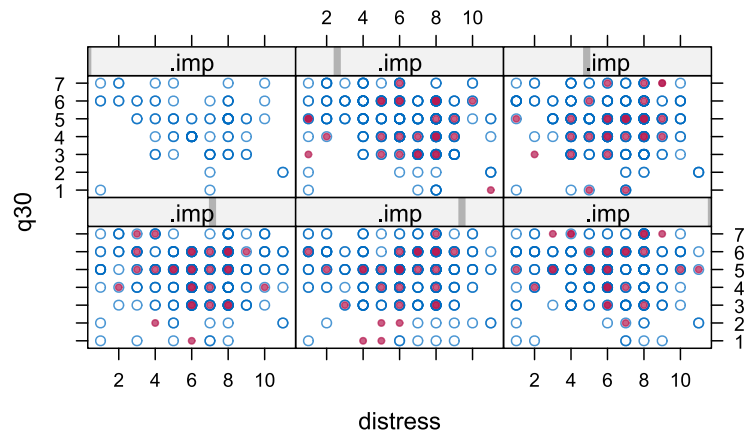


Figure 8: Scatter plot of QoL against distress thermometer score by imputed dataset.

Analysis of Trial Data

For the final trial analysis, all $m = 50$ imputed datasets will be analysed, and posterior draws will be stacked to propagate imputation uncertainty into the final inference.

Simulation Study Dataset

For simulation studies, a single completed dataset is extracted. This is computationally necessary but does not reflect the full imputation uncertainty.

```
complete_data <- complete(imputed_data, 4)

complete_data |>
  mutate(q30 = as.numeric(q30)) |>
  ggplot(aes(x = week, y = q30)) +
  geom_point(alpha = 0.5) +
  labs(x = "Week", y = "QoL (Q30)")
```

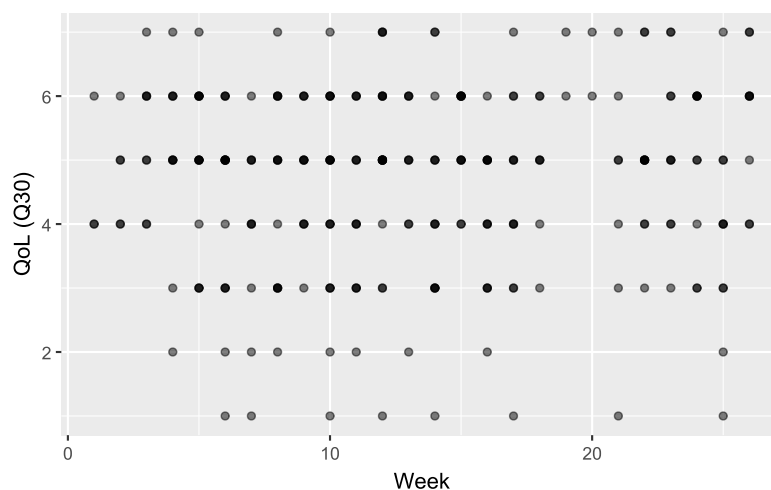


Figure 9: Distribution of QoL scores over time in the selected completed dataset.

Export

```
write_parquet(complete_data, sink = here("data", "qol_complete.parquet"))
```